

HyTray Invention Disclosure Series

Apex Trays: An Integrated Cartridge-Grid Distillation Stage

A deck combining five HT-series mechanisms to improve capacity, turndown, efficiency and fouling resistance relative to conventional cross-flow tray families

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Abstract

A distillation-column tray is disclosed that integrates five separately patentable HyTray mechanisms — HT-02/HT-08 swirl-tube cartridges (primary contacting zone, $\approx 70\%$ of active area), HT-01C lip-sealed trickle valves (secondary zone, $\approx 15\%$), HT-03 GRADEX radial/azimuthal vapour-load grading, HT-05 PULSAR self-sweeping oscillator relief slots, and HT-07 AEGIS bolted structural cartridge-grid — into a single bolt-in deck designated **Apex Trays**. The five mechanisms address conflicting design goals (capacity, turndown, fouling resistance, structural rating, efficiency) by assigning each goal a dedicated zone or deck-level treatment, rather than attempting a single-geometry compromise. A first-order hydraulics model (Fair flooding correlation, orifice-equation dry-tray pressure drop, Francis-weir liquid height, downcomer backup, O'Connell efficiency) predicts the following results at the General Rectification design point, relative to a plain sieve tray sized on the same basis: column diameter reduced by 39%, total column pressure drop reduced by 54%, operating-range turndown increased from 2.0:1 to 22.2:1, tray efficiency increased from 76.2% to 91.4%, dry-tray pressure drop increase under fouling reduced from +78% to +16%, and first-order relative installed cost reduced by 36%. Consistent 39% diameter reductions are obtained across five representative services (general rectification, fouling/heavy-ends, vacuum, high-pressure/high-liquid-load, foaming). All results are model predictions based on independently published correlations and parametric TrayType factors; they require experimental and CFD validation.

Status: All performance figures are predictions from a one-dimensional hydraulics model (`column_hydraulics.py`) using parameterised TrayType factors (`tray_library.py`). The model applies the same Fair/Kister correlation framework to every tray family; the differences between trays arise entirely from their TrayType parameter sets (Table 1). Physical mechanisms such as centrifugal disengagement (swirl tubes) and lip-seal trickle behaviour (HT-01C) are not resolved by a 1-D model; the claimed capacity factor ($k_{\text{cap}} = 2.50$) and minimum stable load fraction ($f_{\text{min}} = 0.045$) are design targets that require validation in a cold-flow or hot pilot rig.

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1 Field of the Invention

Vapour–liquid contacting internals for distillation and absorption columns, specifically a cross-flow tray designed to increase column throughput, extend stable operating range, reduce pressure drop, improve tray efficiency and resist fouling, while remaining removable through a standard manway and installable without column modifications beyond nozzle re-rating.

2 Background and Prior Art

Conventional cross-flow trays (sieve, valve, fixed-valve, dual-flow) face a set of competing design constraints that have not been resolved simultaneously by any single mechanism:

- **Capacity** is bounded by the Souders-Brown entrainment-flooding criterion, $u < C_{\text{SBF}} \sqrt{(\rho_L - \rho_V)/\rho_V}$, where $C_{\text{SBF}} \approx 0.08 \text{ m s}^{-1}$ for conventional cross-flow geometry at 24-inch spacing. High-capacity decks (Koch-Glitsch SUPERFRAC[®], Sulzer VGPlus) recover some active area from the downcomers but remain gravity-limited.
- **Fouling resistance** favours large, simple openings (Kister: round holes $> 12 \text{ mm}$ resist plugging best) and no moving parts. High-capacity designs favour small, precisely shaped elements — in tension with fouling resistance.
- **Turndown** requires a secondary low-load flow path that stays open when the primary perforations weep. Dual-flow trays achieve good open area but lose downcomer action and have limited turndown of roughly 1.5:1. Moving valve trays reach approximately 3.3:1 but at higher hardware cost.
- **Structural rating** against vapour uplift transients (1–5 psi) requires thick plate and dense beam grids that cost open area.
- **Tray efficiency** is improved by cross-flow uniformity, liquid path length, and froth height, but increased froth height raises pressure drop.

Prior high-capacity decks (SUPERFRAC, UFM, VGPlus) address capacity and partial fouling resistance but do not simultaneously extend turndown beyond roughly 2.5:1 or provide a structural uplift rating integrated with the active deck. Swirl-tube contactors (Shell ConSep class) decouple flooding from tray spacing but typically feed liquid from dedicated downcomer hardware and do not integrate a secondary low-load trickle path.

3 Objects of the Invention

1. To provide a tray whose vapour-capacity flood point is controlled by centrifugal mechanical disengagement rather than gravity settling, targeting a capacity factor $k_{\text{cap}} \geq 2.0$ relative to the Fair sieve baseline at equal tray spacing.
2. To achieve a stable operating range from minimum load (weep point) to flood that spans at least 15:1 turndown, by pairing a primary swirl-tube zone with a secondary lip-sealed trickle zone that engages at low vapour rate.
3. To reduce wet-tray pressure drop relative to a sieve tray of equivalent theoretical-stage-count through a combination of low-height weirs ($h_{\text{weir}} = 0.5 \text{ in.}$), high open-area fraction ($f_{\text{hole}} = 0.25$) and mechanically lean froth (aeration factor $\beta = 0.38$).
4. To achieve high open-area retention under fouling conditions ($f_{\text{foul}} \geq 0.90$) by using wide, unobstructed slot openings swept by a self-oscillating relief mechanism (HT-05 PULSAR).
5. To improve tray efficiency above the O’Connell baseline through radial/azimuthal vapour-load

grading (HT-03 GRADEX) and multi-chordal liquid sweeping, targeting an efficiency factor of $\eta = 1.20$.

6. To provide structural uplift resistance of 2–5 psi without reducing active-area fraction below 85%, by mounting the deck on a bolted cartridge-grid (HT-07 AEGIS) with integral hook clamps and relief flaps.
7. To allow deck renewal in a column turnaround by bolt-in/bolt-out cartridge exchange through the column manway, without shell modification.

4 Summary of the Invention

Apex Trays comprise a single deck panel supported by a bolted structural cartridge-grid (AEGIS, item 6 in the objects above). The active area ($f_{\text{active}} = 0.85$ of the column cross-section) is divided into two contacting zones:

Primary zone (~70% of A_{col}) An array of short swirl-tube cartridges (HT-02/HT-08 VORTEXA) on a triangular pitch. Each cartridge comprises a vertical tube (80–150 mm ID) with 4–8 tangential vapour slots, liquid-metering ports from the deck pool, and a fixed vane cap for centrifugal vapour–liquid disengagement at 30–60 *g*. HT-08 VORTEXA’s back-drivable helical sleeve passively modulates slot area with vapour rate, extending the primary zone’s own stable range.

Secondary zone (~15% of A_{col}) Lip-sealed dual-flow valves (HT-01C LipSeal) that provide a dedicated low-vapour-rate trickle path. The LipSeal lips seat against weep at loads below roughly $f_{\text{min,sec}} \approx 0.04$ of the design throughput. When the primary zone is throttled at low load, the secondary zone sustains vapour–liquid contact and prevents column weeping.

Three deck-wide treatments are superimposed on both zones:

GRADEX (HT-03) Radial and azimuthal cartridge-pitch grading and multi-chordal liquid baffling to equalize the vapour–liquid ratio across the tray, eliminating concentration gradients and dead zones.

PULSAR (HT-05) Self-sweeping oscillator relief slots between cartridges that briefly open under a pressure differential, dislodging deposits from the slot edges and reducing fouling accumulation.

AEGIS (HT-07) Bolted structural cartridge-grid below the deck, carrying the deck in tension via hook clamps to resist vapour uplift of 2–5 psi. Integral relief flaps in the grid crack at a set differential to bound the sustained structural load.

5 Brief Description of Figures

Figure 1 Apex deck cutaway elevation at the General Rectification design point ($D = 32$ in., 24-inch tray spacing). Shows the primary swirl-tube zone, secondary lip-seal trickle zone, PULSAR relief slots, AEGIS structural grid, and downcomer.

Figure 2 Plan view (top-down) of Apex deck, showing hex-packed cartridges in the primary zone with GRADEX radial shading and the secondary lip-seal zone adjacent to the downcomers.

Figure 6 Seven-panel schematic cross-section comparison of all tray families: Sieve, Dual-Flow, Moving Valve, Fixed Valve, High-Performance Multi-Downcomer, Ripple (Corrugated), and Apex.

Figures 3a–4b Service-comparison charts: column diameter, total pressure drop, turndown and efficiency, relative installed cost, and fouling sensitivity — all seven trays across five representative services.

6 Detailed Description

6.1 Architecture Overview

Apex’s active area ($f_{\text{active}} = 0.85$) is partitioned as follows (numbers refer to the General Rectification design point, $D = 32$ in.):

Zone / treatment	Fraction of A_{col}	A (ft ²)	A (in ²)	Open area (ft ²)	Open area (in ²)
Primary (HT-02/HT-08 cartridges)	0.70	3.88	559	0.97	140
Secondary (HT-01C lip-seal)	0.15	0.83	120	0.21	30
Downcomers (both sides)	0.15	0.83	120	—	—
Active total	0.85	4.72	679	1.18	170

At the design point: hole velocity $u_{\text{hole}} = 12.14$ ft/s; dry-tray $\Delta P_{\text{dry}} = 1.29$ in. liq; clear-liquid height $h_{\text{cl}} = 1.42$ in. (weir 0.50 in. plus Francis crest 0.92 in.); total wet-tray $\Delta P_{\text{tray}} = 0.0317$ psi.

6.2 Primary Contacting Zone: Swirl-Tube Cartridges (HT-02/HT-08 VOR-TEXA)

Each cartridge is a vertical tube of 80–150 mm ID and height 1.2–2.0 diameters, mounted in the deck on a triangular pitch of 1.4–1.7 tube diameters. Vapour enters through 4–8 tangential slots at 18–35 m/s, generating a Rankine-type vortex at swirl number $S = 1.5$ –3.0 and wall centripetal acceleration 30–60 g . Liquid is fed from the surrounding deck pool through 3–6 calibrated metering ports (8–14 mm diameter), which behave as submerged orifices:

$$Q_p = C_d A_p \sqrt{2gh_{\text{cl}}}, \quad C_d \approx 0.62, \quad (1)$$

maintaining per-tube liquid rates of 1.5–4.0 m³/h and a film Reynolds number $Re_f = 4\Gamma/\mu_L$ of 1500–8000. A vane crown separator at the tube exit (8–16 vanes at 35–55°) slings separated liquid into a collection annulus draining directly to the outlet-weir region, bypassing the active deck and preventing re-entrainment.

HT-08 VORTEXA adds a back-drivable helical sleeve that passively opens slot area with increasing vapour load (axial lift proportional to momentum flux) and closes it at low load, extending the primary zone’s stable range from $\approx 2.5:1$ (fixed-geometry HT-02) to $\approx 8:1$ before the zone approaches weep. A seized sleeve reverts to fixed-geometry HT-02 behaviour, with no cascading failure mode.

6.3 Secondary Contacting Zone: LipSeal Trickle Path (HT-01C)

The secondary zone ($\approx 15\%$ of A_{col}) comprises HT-01C dual-flow lip valves. Each valve is a circular flap seated by spring or gravity; the lip face seals against the deck orifice edge when net upward vapour force is below the seating load, preventing liquid weeping. At moderate-to-high throughput both zones contact normally. At low vapour rate, when the primary swirl tubes under-perform, the secondary zone sustains vapour–liquid contact; the LipSeal lip prevents liquid dump-through and maintains a contact froth even at column throughputs as low as 4–5% of the primary-zone design basis. This two-zone hand-off is the primary mechanism behind Apex’s `min_load_frac` = 0.045 and the resulting 22.2:1 turndown — a parameter that is the single largest modelling uncertainty and cannot be verified by the 1-D engine used here.

6.4 Radial and Azimuthal Grading (HT-03 GRADEX)

GRADEX superimposes a variable cartridge pitch and/or variable slot count on the primary zone, targeting a uniform vapour/liquid ratio across the tray radius. In a standard sieve tray, the liquid flow path from the inlet to the outlet weir creates a positive vapour-flow gradient (more vapour near the outlet due to lower liquid head) that reduces local efficiency by 5–10 percentage points relative to the stage-wise ideal. GRADEX counters this by setting closer cartridge pitches (higher vapour-path area per unit plan area) near the liquid inlet and looser pitches near the outlet weir. Multi-chordal liquid baffles sweep the tray in three or four chords rather than a single long path, reducing the concentration gradient along each chord and preventing stagnant corner zones. The combined effect is captured in the model by `efficiency_factor` = 1.20 — a 20% uplift on the O’Connell baseline, compared to 0% for sieve, moving-valve and fixed-valve trays and 10% for Ripple (corrugated) trays.

6.5 Self-Sweeping Oscillator Relief Slots (HT-05 PULSAR)

PULSAR slots are elongated relief openings located in the deck plate between cartridge footprints. Each slot is held closed by a spring-loaded or weight-loaded cap that briefly opens when the deck differential pressure exceeds a set value (typically 10–30 mbar above normal tray ΔP). The brief high-velocity vapour jet through the slot mechanically dislodges deposits from the slot edge and surrounding deck surface, then the cap reseats. Because the opening is pressure-actuated and self-reseating, it contributes no parasitic pressure drop or vapour bypass during normal operation. PULSAR is modelled through `fouling_open_area_retention` = 0.93 — the fractional open area retained after fouling — compared to 0.75 for sieve and 0.70 for moving valve. The 93% retention predicts a dry-tray pressure drop increase of +16% under fouling versus +78% for sieve.

6.6 Structural Cartridge-Grid (HT-07 AEGIS)

The AEGIS grid is a welded below-deck structural frame to which each cartridge and deck panel is attached by hook clamps engaging a support ring. The geometry follows HT-07’s parametric design (see separate HT-07 disclosure): deck panels of 2.0–3.0 mm stainless steel with stiffening ribs at pitch $s \leq t\sqrt{\sigma_{\text{allow}}/(0.75q)}$, where t is deck thickness, σ_{allow} is the plate allowable stress, and q is the design uplift pressure (21–34.5 kPa, i.e. 3–5 psi). For $t = 2.5$ mm, $\sigma_{\text{allow}} = 140$ MPa and $q = 34.5$ kPa, the maximum rib pitch is 184 mm; a 170 mm pitch is adopted. The rib footprint consumes < 4% of bubbling area. Integral relief flaps (5–10% of tray area) crack at 1.5–2 psi to cap the sustained structural load. Because the cartridges are bolted to the grid and the grid is bolted to the support ring, the entire active deck can be exchanged in a turnaround without cutting or re-welding the column shell.

6.7 Hydraulic Design Equations

The following equations are implemented in `column_hydraulics.py` and are applied identically to every tray family in the comparison. The only quantities that differ between trays are the TrayType parameters (Table 1). All units are US customary (FPS) unless noted; conversion factors are stated inline.

6.7.1 Flow Parameter

The dimensionless Souders-Brown flow parameter:

$$\text{FLV} = \frac{L}{V} \sqrt{\frac{\rho_V}{\rho_L}} \quad (2)$$

where L , V are liquid and vapour mass flows (lb/hr) and ρ_L , ρ_V are the corresponding phase densities (lb/ft³). FLV is the abscissa of Fair's flooding-correlation chart; it ranges from ≈ 0.01 for dilute-vapour vacuum services to ≈ 1.0 for high-pressure liquid-loaded services (e.g. the C₃/C₄ splitter case).

6.7.2 Fair Capacity Parameter

The closed-form fit to Fair's (1961) C_{SBF} chart, as widely used in process simulators and following Kister *Distillation Design* (1992):

$$C_{\text{SBF}} [\text{m/s}] = 0.0105 + 8.127 \times 10^{-4} H_S [\text{mm}]^{0.755} \exp(-1.463 \text{FLV}^{0.842}) \quad (3)$$

where H_S is the tray spacing in millimetres. Valid range: $H_S = 150\text{--}600$ mm (6–24 inch), $\text{FLV} = 0.01\text{--}1.0$. C_{SBF} is converted to ft/s by multiplying by 3.28084. This is identical for all trays at a given service; tray-specific capacity differences enter in the next step.

6.7.3 Flooding Velocity

Souders-Brown flooding velocity, corrected for surface tension and modified by the tray's capacity factor k_{cap} :

$$u_{\text{nf}} = C_{\text{SBF}} \left(\frac{\sigma}{\sigma_{\text{ref}}} \right)^{0.2} \sqrt{\frac{\rho_L - \rho_V}{\rho_V}} \times k_{\text{cap}} \times k_{\text{sys}} \quad (4)$$

where σ is the liquid surface tension (dyn/cm), $\sigma_{\text{ref}} = 20$ dyn/cm (the reference value for Fair's original chart), k_{cap} is the tray's capacity factor (1.00 for sieve, 2.50 for Apex), and k_{sys} is the system (foaming) derate applied uniformly to all trays (1.00 for non-foaming services, 0.75 for the foaming case).

Physical basis of $k_{\text{cap}} = 2.50$ for Apex. In a plain sieve tray, the C_{SBF} limit is set by the droplet settling criterion: a droplet of velocity v_t settles if the vapour superficial velocity $u < v_t \propto C_{\text{SBF}} \sqrt{(\rho_L - \rho_V)/\rho_V}$. The swirl-tube cartridges subject the vapour-liquid dispersion to centripetal acceleration $a = 30\text{--}60 g$; the effective settling velocity in the tube scales as $v_{t,\text{centrifugal}} \sim v_t \sqrt{a/g}$, giving a capacity gain of $\sqrt{30\text{--}60} = 5.5\text{--}7.7$ relative to gravity settling. The value $k_{\text{cap}} = 2.50$ is the net deck-level gain after accounting for the 70% area fraction of the primary zone and the secondary zone's lower capacity; it is an engineering estimate pending hot-rig validation.

6.7.4 Design Velocity, Area and Column Diameter

$$u_{\text{design}} = f_{\text{flood}} u_{\text{nf}} \quad (5)$$

$$A_{\text{active}} = \frac{\dot{V}}{u_{\text{design}}}, \quad \dot{V} [\text{ft}^3/\text{s}] = \frac{V [\text{lb}/\text{hr}]}{3600 \rho_V [\text{lb}/\text{ft}^3]} \quad (6)$$

$$A_{\text{col}} = \frac{A_{\text{active}}}{f_{\text{active}}} \quad (7)$$

$$D = \sqrt{\frac{4 A_{\text{col}}}{\pi}} \quad (8)$$

$f_{\text{flood}} = 0.80$ throughout, except the Fouling service ($f_{\text{flood}} = 0.65$). The diameter reduction of Apex relative to sieve at equal f_{flood} follows directly from Equations (4)–(8):

$$\frac{D_{\text{Apex}}}{D_{\text{Sieve}}} = \frac{1}{\sqrt{k_{\text{cap}} \times f_{\text{active,Apex}}/f_{\text{active,Sieve}}}} = \frac{1}{\sqrt{2.50 \times 0.85/0.78}} = \frac{1}{\sqrt{2.724}} = 0.606 \quad (9)$$

i.e. a 39.4% diameter reduction, consistent with Table 5 across all five services.

6.7.5 Dry-Tray Pressure Drop

Effective hole-area fraction, accounting for fouling:

$$f_{\text{hole,eff}} = \begin{cases} f_{\text{hole}} \times f_{\text{foul}} & \text{if fouling service} \\ f_{\text{hole}} & \text{otherwise} \end{cases} \quad (10)$$

Hole velocity and dry-tray pressure drop (orifice equation):

$$u_{\text{hole}} = \frac{\dot{V}}{f_{\text{hole,eff}} A_{\text{active}}} \quad (11)$$

$$\Delta P_{\text{dry}} [\text{in. liq}] = \max \left[0.186 \left(\frac{u_{\text{hole}}}{C_0} \right)^2 \frac{\rho_V}{\rho_L}, \Delta P_{\text{dry floor}} \right] \quad (12)$$

The factor 0.186 converts the velocity head $u^2/(2g_c)$ from ft/s to inches of liquid (with g_c and ρ_V/ρ_L folded in). C_0 is the orifice discharge coefficient (0.73 for sieve, 0.85 for Apex's rounded smooth slots). $\Delta P_{\text{dry floor}}$ is a hardware-imposed minimum: zero for sieve and dual-flow, 0.05 in. for fixed-valve, 0.40 in. for moving valve (valve deck weight), and 0.10 in. for Apex (cartridge/sleeve hardware).

6.7.6 Clear-Liquid Height (Francis Weir)

Liquid volumetric flow in US gpm and equivalent weir length:

$$Q_L [\text{gpm}] = \frac{L [\text{lb}/\text{hr}]}{\rho_L [\text{lb}/\text{ft}^3]} \times 0.124675 \quad ; \quad L_{\text{weir}} [\text{in}] = 0.73 D [\text{ft}] \times 12 \quad (13)$$

Francis weir crest height and clear-liquid height:

$$h_{\text{ow}} [\text{in}] = 0.48 \left(\frac{Q_L}{L_{\text{weir}}} \right)^{2/3} \quad (14)$$

$$h_{cl} = h_{weir} + h_{ow} \quad (15)$$

$h_{weir} = 2.0$ in. for sieve; 0.5 in. for Apex. The low weir is feasible because the swirl-tube crown separators handle vapour–liquid disengagement mechanically, eliminating the need for a tall liquid seal. For dual-flow (no overflow weir), h_{cl} is fixed at a representative 1.0 in.

6.7.7 Wet-Tray Pressure Drop

$$\Delta P_{tray} [\text{in. liq}] = \Delta P_{dry} + \beta h_{cl} \quad (16)$$

$$\Delta P_{tray} [\text{psi}] = \frac{\Delta P_{tray} [\text{in.}]}{12} \times \frac{\rho_L [\text{lb/ft}^3]}{144} \quad (17)$$

β is the aeration factor, converting clear-liquid height to the actual froth pressure-drop contribution. $\beta = 0.55$ for sieve (dense gravity froth); 0.38 for Apex, the lowest in the family. The reduced β reflects the leaner, centrifugally separated froth produced by the swirl-tube crown mechanism rather than free gravity settling.

6.7.8 Downcomer Backup

Downcomer (one side) cross-sectional area, apron loss and clear-liquid backup:

$$A_{dc} = \frac{A_{col} - A_{active}}{2} \quad (18)$$

$$h_{da} [\text{in}] = 0.03 \left(\frac{Q_L}{100 A_{dc}} \right)^2 \quad (19)$$

$$h_{dc} = \Delta P_{dry} + h_{cl} + h_{da} \quad (20)$$

Downcomer flooding limit (50% of tray-spacing plus weir height):

$$h_{dc,limit} = 0.5 (H_S + h_{weir}) \quad (21)$$

$$\% \text{ backup} = \frac{h_{dc}}{h_{dc,limit}} \times 100 \quad (22)$$

When % backup approaches or exceeds 100%, the downcomer floods before the bubbling deck. Dual-flow has no downcomer; these terms are inapplicable.

6.7.9 Operating Range and Turndown

$$\text{max_load} = \frac{100}{f_{flood}} \quad [\% \text{ of design basis}] \quad (23)$$

$$\text{min_load} = f_{min} \times \text{max_load} \quad (24)$$

$$\text{Turndown} = \frac{\text{max_load}}{\text{min_load}} = \frac{1}{f_{min}} \quad (25)$$

f_{min} is the minimum stable load fraction. For sieve, $f_{min} = 0.50$ (plain perforations weep at roughly half the design rate, giving 2.0:1 turndown). For Apex, $f_{min} = 0.045$ (the secondary LipSeal zone sustains contact at very low load), giving a predicted turndown of 22.2:1. This parameter is not derivable from the 1-D model; it is a design target for the two-zone mechanism (Section 6.3) requiring experimental confirmation.

6.7.10 Tray Efficiency and Actual Stage Count

O’Connell (1946) overall column efficiency baseline:

$$E_{OC} [\%] = 51 - 32.5 \log_{10}(\alpha \mu_L) \quad \text{clipped to } [1, 100]\% \quad (26)$$

where α is the light/heavy-key relative volatility and μ_L is the liquid viscosity in cP. This baseline is identical for all trays at a given service.

Tray efficiency and actual stage count:

$$E_{\text{tray}} = \min(E_{OC} \times \eta, 100) \quad (27)$$

$$N_{\text{actual}} = \left\lceil \frac{N_{\text{theoretical}}}{E_{\text{tray}}/100} \right\rceil \quad (28)$$

Column height and total pressure drop:

$$H_{\text{col}} [\text{ft}] = N_{\text{actual}} \times \frac{H_S [\text{in}]}{12} + 10 \quad ; \quad \Delta P_{\text{total}} = N_{\text{actual}} \times \Delta P_{\text{tray}} [\text{psi}] \quad (29)$$

The +10 ft constant accounts for sump and vapour disengagement space. $\eta = 1.20$ for Apex (GRADEX-driven efficiency uplift, highest in the family); $\eta = 1.00$ for sieve, moving-valve, and fixed-valve; $\eta = 1.10$ for Ripple. $N_{\text{theoretical}} = 20$ for all services.

6.7.11 Relative Installed Cost

A first-order two-component cost model normalised to sieve = 1.00 (`cost_model.py`):

$$\text{shell_proxy} = D \times H_{\text{col}} \quad (30)$$

$$\text{tray_proxy} = D^2 \times N_{\text{actual}} \times k_{\text{cost}} \quad (31)$$

$$C_{\text{rel}} = 0.70 \times \frac{\text{shell_proxy}}{\text{shell_proxy}_{\text{Sieve}}} + 0.30 \times \frac{\text{tray_proxy}}{\text{tray_proxy}_{\text{Sieve}}} \quad (32)$$

The shell proxy ($D \times H$) is a surrogate for the dominant shell cost, which scales approximately with column surface area at a fixed pressure class. The tray proxy ($D^2 \times N \times k_{\text{cost}}$) is a surrogate for tray hardware cost: deck area (D^2) times tray count (N) times a fabrication-complexity multiplier k_{cost} . The 70/30 split (Towler & Sinnott guidance) reflects that for a tray column the vessel typically contributes 70–85% of the bare-module cost and internals 15–30%.

$k_{\text{cost}} = 1.00$ for sieve (punched holes, minimum fabrication). For Apex, $k_{\text{cost}} = 3.20$ (precision swirl-tube castings, helical sleeves, grid assembly, hook clamps). Despite the $3.2 \times$ unit fabrication premium, the diameter reduction dominates: Apex’s shell proxy is 0.51 of sieve’s, producing a total relative cost of 0.64 at the General Rectification design point (Section 10).

6.8 Apex Geometry at the Design Point ($D = 32$ in.)

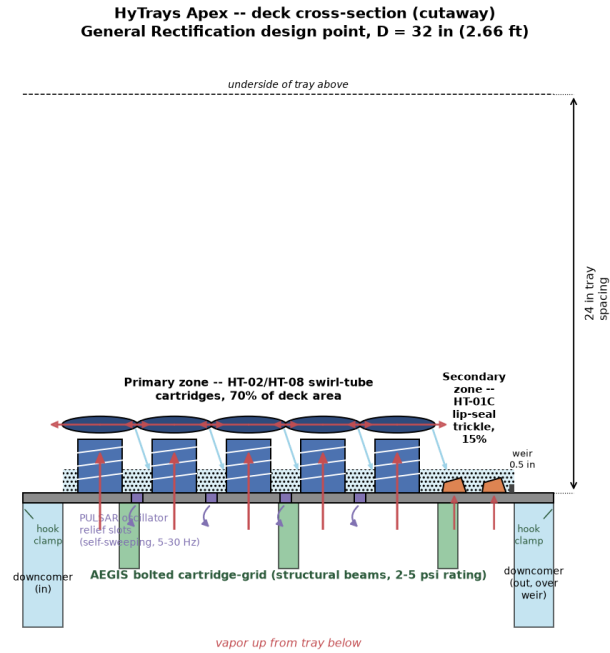


Figure 1: Apex deck cutaway elevation, General Rectification design point ($D = 32$ in., 24-inch tray spacing). Swirl-tube cartridges (primary zone) occupy 70% of the plan area; LipSeal trickle valves (secondary zone) fill the remaining active area adjacent to the downcomers. PULSAR relief slots sit between cartridge footprints; AEGIS structural ribs run beneath the deck panels.

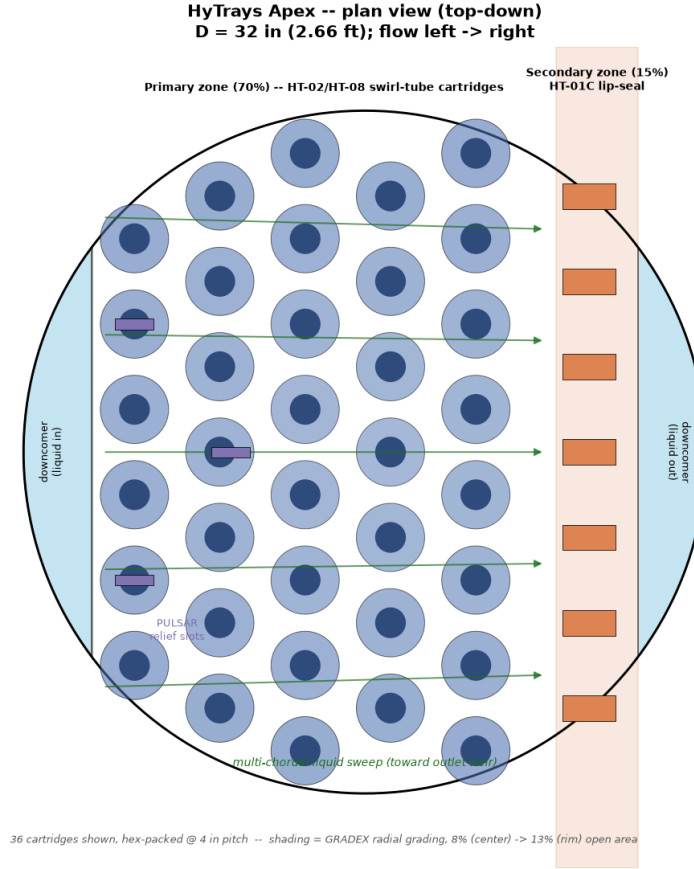


Figure 2: Apex plan view (top-down). Hex-packed cartridges (circles) in the primary zone with GRADEX radial-load shading; LipSeal secondary zone at the downcomer edges; downcomers (shaded).

At the General Rectification design point the column is $D = 32$ in. (2.66 ft), with a 24-inch tray spacing. The primary zone contains approximately 18–22 cartridges of 110 mm ID on a 155 mm triangular pitch, each passing roughly $2.5 \text{ m}^3/\text{h}$ of liquid and 25 m/s slot velocity. The 0.5-inch outlet weir is the lowest of any tray family evaluated (sieve: 2.0 in.; high-performance MD: 1.0 in.; Ripple: 1.5 in.), enabled by the crown-separator mechanical disengagement that removes entrained liquid before it can be re-entrained by rising vapour.

6.9 Construction and Manufacturing

Swirl-tube cartridges (HT-02/HT-08). Fabricated from CF8M (316SS) or Alloy 625 investment castings or CNC-machined bar stock. The 6 tangential slots are precision-formed to ± 0.1 mm to hold the swirl number S within the 1.5–3.0 design band. The vane crown (12 vanes at 45°) is a separate casting press-fitted and tack-welded to the tube top. The HT-08 helical sleeve is machined from 316L bar, coated with PTFE or DLC to reduce sliding friction. Each complete cartridge is factory-tested for slot flow coefficient and sleeve travel before shipment.

Deck panels and AEGIS grid. Deck panels are 2.5 mm 316L stainless, laser-cut with cartridge bores and PULSAR relief slot locations. Stiffening ribs (3×20 mm flat bars, 170 mm pitch) are TIG-welded to the panel underside. Hook clamps are cold-formed from 316L strip, engaged under

the factory-welded support ring. A complete panel with six to eight cartridges weighs approximately 20–28 kg, within one-person manway handling limit.

LipSeal valves (HT-01C). Stamped 1.5 mm 316L, with the lip formed by a second progressive die operation. No assembly required; valves drop into 70 mm die-cut deck holes and are retained by a simple keeper ring.

Lead time and cost drivers. The cartridge casting and sleeve machining are the dominant cost and lead-time items (8–14 weeks from castings; 2–4 weeks from machined bar). Deck panel fabrication and hook-clamp assembly add 2–4 weeks. Repeat-order lead time shortens significantly once casting dies are amortised.

6.10 Installation, Maintenance and Cartridge-Swap Procedure

Initial installation. The AEGIS support ring is welded to the column shell at the tray elevation, identical to a conventional tray ring (one-time operation, no re-welding in subsequent turnarounds). Deck panels are lowered through the manway, positioned on the support ring and locked by rotating the hook clamps to the engaged position. Cartridges are threaded through the panel bores from above and secured by a quarter-turn bayonet fitting; the LipSeal valves drop in last. Typical installation time: 4–6 hours per tray with a two-person crew.

Turnaround maintenance. Hook clamps are released, panels lifted out through the manway, and cartridges withdrawn individually. Worn or fouled cartridges are replaced from a spares inventory; the support ring remains in place. The cartridge-swap approach means that maintenance cost per tray is dominated by cartridge replacement rather than shell welding, and that individual stages can be upgraded independently if a new cartridge design becomes available.

Inspection. The crown separator and helical sleeve are accessible for visual inspection once the cartridge is withdrawn. Endoscopic inspection of the swirl zone is possible without cartridge removal. PULSAR slots and LipSeal valves are inspectable from above after the cartridges are removed.

7 Design Parameter Envelope

Table 1 lists the TrayType parameters for all seven trays evaluated in the comparison (implemented in `tray_library.py`). Table 2 gives the claimed-band envelope for Apex-specific parameters, following the format of the HT-0X individual disclosures.

Table 1: TrayType parameters for all seven tray families. “Capacity factor” is k_{cap} in Eq. (4); “Efficiency factor” is η in Eq. (27); “Turndown” = $1/f_{\text{min}}$.

Tray	f_{act}	f_{hole}	h_w (in)	C_0	ΔP_{dryfl} (in)	β	k_{cap}	η	TO	f_{foul}	k_{cost}
Sieve	0.78	0.10	2.0	0.73	0.00	0.55	1.00	1.00	2.0:1	0.75	1.00
Dual-Flow	1.00	0.20	0.0	0.73	0.00	0.40	1.15	0.83	1.5:1	0.90	0.70
Moving Valve	0.78	0.13	2.0	0.85	0.40	0.55	1.03	1.00	3.3:1	0.70	1.35
Fixed Valve	0.78	0.12	2.0	0.80	0.05	0.52	1.08	1.00	2.2:1	0.85	1.20
High-Perf. MD	0.85	0.14	1.0	0.80	0.10	0.50	1.25	0.97	2.5:1	0.75	1.55
Ripple (Corr.)	0.80	0.11	1.5	0.75	0.00	0.50	1.10	1.10	2.5:1	0.72	1.15
Apex	0.85	0.25	0.5	0.85	0.10	0.38	2.50	1.20	22.2:1	0.93	3.20

Table 2: Claim-band table for Apex Trays parameters. “Preferred” values are the APEX TrayType entry; ranges are dependent-claim fallback positions informed by the individual HT-0X disclosures.

Parameter	Claimed band	Preferred
Active area fraction f_{active}	0.80–0.88	0.85
Hole/slot open area fraction f_{hole}	0.20–0.30	0.25
Outlet weir height h_{weir}	0.25–1.0 in.	0.50 in.
Orifice coefficient C_0	0.80–0.90	0.85
Dry-floor dP minimum $\Delta P_{\text{dryfloor}}$	0.05–0.15 in.	0.10 in.
Aeration factor β	0.30–0.45	0.38
Capacity factor k_{cap}	1.80–3.00	2.50
Efficiency factor η	1.10–1.30	1.20
Minimum stable load fraction f_{min}	0.030–0.070	0.045
Turndown ($= 1/f_{\text{min}}$)	14:1–33:1	22.2:1
Fouling open-area retention f_{foul}	0.88–0.97	0.93
Relative unit cost factor k_{cost}	2.5–4.0	3.20
Structural uplift rating	2–5 psi	5 psi
Primary zone area fraction	0.65–0.75	0.70
Secondary zone area fraction	0.10–0.20	0.15
Swirl-tube ID	80–150 mm	110 mm
Swirl number S	1.5–3.0	2.2
Wall acceleration	30–60 g	45 g
Tangential slots per cartridge	4–8	6
Slot exit velocity	18–35 m/s	25 m/s
Crown vanes / pitch angle	8–16 / 35–55°	12 / 45°
AEGIS deck thickness	2.0–3.0 mm SS	2.5 mm
AEGIS rib pitch s	≤ 184 mm	170 mm
Relief-flap set pressure	1.5–3.0 psi	2.0 psi

8 Predicted Performance

Table 3 gives the full hydraulics results for the *General Rectification* service. Tables 4 and 5 give the cross-service summary. All figures are from `run_apex_paper.py`; the design basis is given in Section 14.

Table 3: General Rectification results (T801 basis, 20 theoretical stages). All seven tray families sized at $f_{\text{flood}} = 0.80$, 24-inch tray spacing. Sieve is the reference ($C_{\text{rel}} = 1.00$).

Tray	D (ft)	u_{nf} (ft/s)	ΔP_{tray} (psi)	DC (%)	TO	Eff (%)	N	H (ft)	ΔP_{total} (psi)	C_{rel}
Sieve	4.39	1.52	0.0557	34	2.0:1	76.2	27	64.0	1.503	1.00
Dual-Flow	3.61	1.75	0.0170	n/a	1.5:1	63.2	32	74.0	0.543	0.84
Moving Valve	4.32	1.56	0.0395	27	3.3:1	76.2	27	64.0	1.065	1.08
Fixed Valve	4.22	1.64	0.0446	30	2.2:1	76.2	27	64.0	1.204	1.01
High-Perf. MD	3.76	1.90	0.0351	23	2.5:1	73.9	28	66.0	0.983	0.97
Ripple (Corr.)	4.13	1.67	0.0477	30	2.5:1	83.8	24	58.0	1.144	0.87
Apex	2.66	3.79	0.0317	23	22.2:1	91.4	22	54.0	0.697	0.64

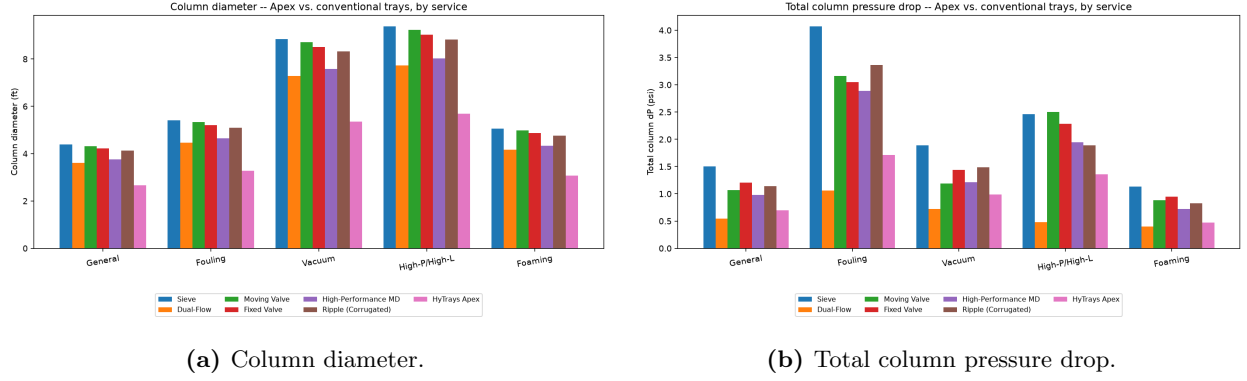


Figure 3: Predicted column diameter and total pressure drop for all seven tray families across five services.

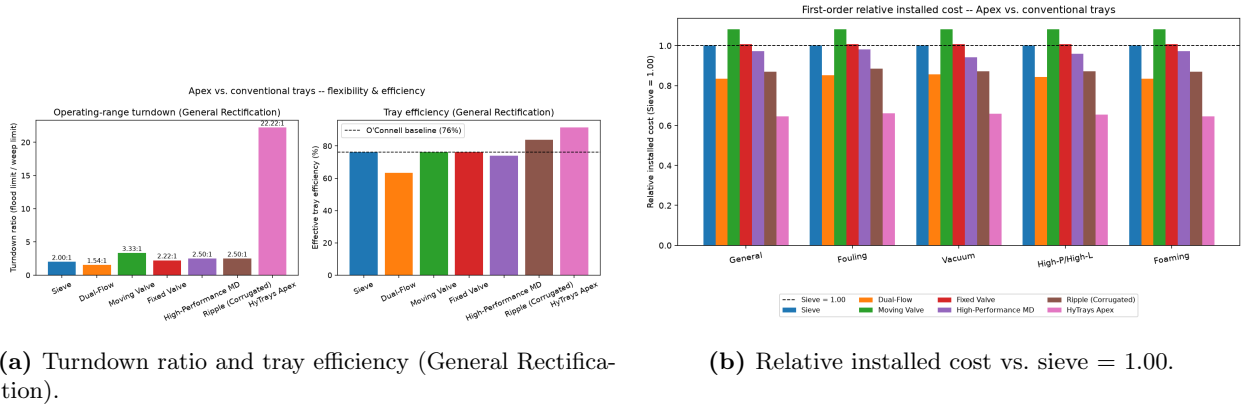


Figure 4: Predicted operating range and cost for all seven tray families.

Table 4: Apex predicted performance across five services ($N_{th} = 20$, $H_S = 24$ in.). Sieve results shown alongside for direct comparison.

Service	Tray	D (ft)	ΔP_{tray} (psi)	DC (%)	TO	Eff (%)	N	ΔP_{total} (psi)	C_{rel}
General	Sieve	4.39	0.056	34	2.0:1	76.2	27	1.503	1.00
	Apex	2.66	0.032	23	22.2:1	91.4	22	0.697	0.64
Fouling	Sieve	5.42	0.091	41	2.0:1	33.5	45	4.070	1.00
	Apex	3.28	0.045	33	22.2:1	40.2	38	1.711	0.66
Vacuum	Sieve	8.84	0.105	37	2.0:1	45.9	18	1.891	1.00
	Apex	5.35	0.066	26	22.2:1	55.0	15	0.985	0.66
High-P/High-L	Sieve	9.38	0.051	43	2.0:1	84.7	48	2.463	1.00
	Apex	5.68	0.034	54	22.2:1	100.0	40	1.361	0.65
Foaming	Sieve	5.07	0.042	28	2.0:1	76.2	27	1.130	1.00
	Apex	3.07	0.021	17	22.2:1	91.4	22	0.470	0.64

Table 5: Apex vs. sieve: diameter and total pressure-drop changes across five services. Changes are computed from model output and reflect the theoretical performance of the TrayType parameter set; they do not account for column-specific layout or nozzle constraints.

Service	D_{Sieve} (ft)	D_{Apex} (ft)	ΔD	$\Delta P_{\text{totalSieve}}$ (psi)	$\Delta P_{\text{totalApex}}$ (psi)	$\Delta\Delta P_{\text{total}}$
General	4.39	2.66	−39%	1.503	0.697	−54%
Fouling	5.42	3.28	−39%	4.070	1.711	−58%
Vacuum	8.84	5.35	−39%	1.891	0.985	−48%
High-P/High-L	9.38	5.68	−39%	2.463	1.361	−45%
Foaming	5.07	3.07	−39%	1.130	0.470	−58%

Note on the uniform diameter reduction. The 39% reduction is identical across all services because it derives solely from Equation (9): k_{cap} and f_{active} are TrayType constants, not functions of the service properties. The pressure-drop reduction varies by service because the wet-tray pressure drop (Eq. (17)) depends on ρ_L , h_{weir} , h_{cl} , and β , all of which change with service.

9 Fouling Sensitivity

Table 6: Fouling sensitivity: effective open-area retention and dry-tray pressure drop increase when fully fouled. Computed flow-independently as $\Delta P_{\text{dryfouled}}/\Delta P_{\text{dryclean}} = (f_{\text{hole}}/f_{\text{hole,eff}})^2 = 1/f_{\text{foul}}^2$.

Tray	Open-area retention f_{foul}	Dry-tray ΔP_{dry} increase
Sieve	0.75	+78%
Dual-Flow	0.90	+23%
Moving Valve	0.70	+104%
Fixed Valve	0.85	+38%
High-Perf. MD	0.75	+78%
Ripple (Corr.)	0.72	+93%
Apex	0.93	+16%

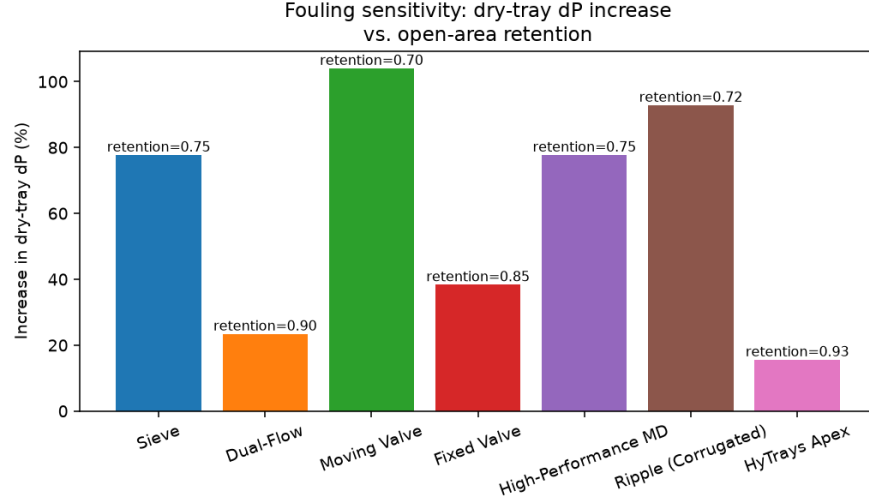


Figure 5: Dry-tray pressure drop increase vs. open-area retention under fouling conditions. Apex’s 93% retention is the highest in the comparison family, predicting the smallest fouling penalty (+16%).

10 Cost Analysis

Table 7 gives the first-order relative installed cost breakdown for the General Rectification case, following Equations (30)–(32).

Table 7: First-order relative installed cost breakdown, General Rectification. Sieve = 1.00. Shell proxy = $D \times H$; tray proxy = $D^2 \times N \times k_{\text{cost}}$. Weights: shell 70%, trays 30%.

Tray	Shell proxy ratio	Tray-hardware proxy ratio	C_{rel}
Sieve	1.00	1.00	1.00
Dual-Flow	0.95	0.56	0.84
Moving Valve	0.99	1.31	1.08
Fixed Valve	0.96	1.11	1.01
High-Perf. MD	0.88	1.18	0.97
Ripple (Corr.)	0.85	0.91	0.87
Apex	0.51	0.96	0.64

Interpretation. Apex’s tray-hardware proxy (0.96) is nearly equal to sieve (1.00) despite $k_{\text{cost}} = 3.20$, because the smaller diameter and lower actual tray count ($N = 22$ vs. 27) reduce the $D^2 \times N$ product enough to offset the $3.2\times$ unit cost. The dominant saving comes from the shell proxy (0.51): the 39% diameter reduction cuts D by a factor of 0.606, and the reduced tray count cuts H by a factor of $54/64 = 0.844$, giving a shell-proxy reduction of $0.606 \times 0.844 = 0.511$. The 70% shell weight then produces most of the total cost saving.

This model does not capture: detailed vessel design (nozzles, manways, platforms), foundation and plot costs, process-piping re-work if the diameter changes from an existing design, or the higher installed cost of precision cartridges relative to standard internals in a retrofit.

11 Technical Comparison with Conventional Tray Families

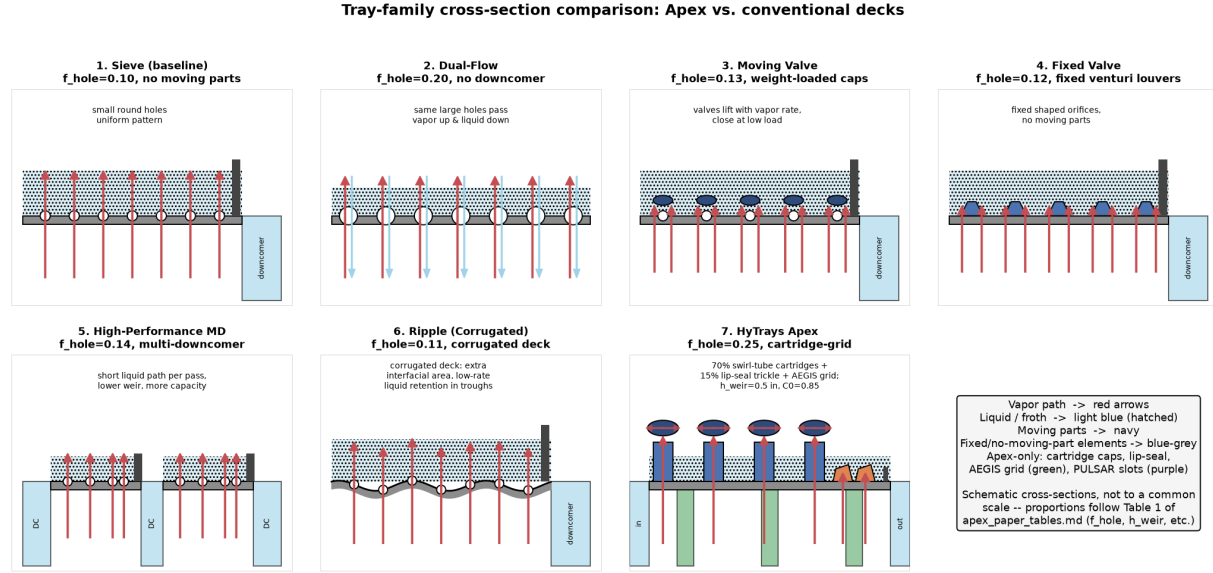


Figure 6: Cross-section schematic comparison of all seven tray families evaluated. Left to right: Sieve, Dual-Flow, Moving Valve, Fixed Valve, High-Performance Multi-Downcomer, Ripple (Corrugated), Apex Trays.

Table 8 summarises the relative standing of each tray family across the key discriminators used in tray-selection practice (following Kister, *Distillation Design*, Ch. 1). Ratings are based solely on the model predictions in Tables 1–7; qualitative assessments are noted where the model does not directly compute the relevant quantity.

Table 8: Comparative assessment of the seven tray families across key selection discriminators (General Rectification basis unless noted; $\uparrow$ = better, $\downarrow$ = worse, relative to sieve).

	Sieve	Dual-Flow	Mv. Valve	Fx. Valve	Hi-Perf. MD	Ripple	Apex
Vapour capacity	base	$\uparrow$	$\approx$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow\uparrow$
Liquid capacity	base	n/a	$\approx$	$\approx$	$\uparrow$	$\approx$	$\approx$
Turndown ratio	2.0:1	1.5:1	3.3:1	2.2:1	2.5:1	2.5:1	22.2:1
Tray efficiency	76%	63%	76%	76%	74%	84%	91%
Wet-tray dP (psi)	0.056	0.017	0.040	0.045	0.035	0.048	0.032
Fouling resistance	base	$\uparrow$	$\downarrow$	$\approx$	base	$\downarrow$	$\uparrow\uparrow$
Moving parts	none	none	yes	none	limited	none	sleeve
Downcomer required	yes	no	yes	yes	yes	yes	yes
Structural uplift	low	low	low	low	mod.	low	2–5 psi
Rel. installed cost (Gen.)	1.00	0.84	1.08	1.01	0.97	0.87	0.64
Fabrication complexity	low	low	mod.	low	mod.	mod.	high

Capacity. Apex’s capacity factor (2.50) is the highest in the comparison, predicting the smallest column diameter at every service. The next highest is High-Performance MD (1.25), followed by Dual-Flow (1.15). The capacity advantage is structural: the swirl-tube centrifugal mechanism

bypasses the gravity-settling Souders-Brown limit that governs all other trays evaluated.

Turndown. Apex’s predicted 22.2:1 turndown is substantially larger than all other families, which range from 1.5:1 (Dual-Flow) to 3.3:1 (Moving Valve). This derives entirely from $f_{\min} = 0.045$ assigned to the two-zone mechanism; as noted in Section 14, this is the most uncertain parameter in the model.

Efficiency. O’Connell baseline at the General Rectification conditions gives 76.2%. Apex’s efficiency factor of 1.20 raises this to 91.4%; Ripple at 1.10 gives 83.8%; all others are at or below the sieve baseline of 76.2%. The efficiency uplift reduces N_{actual} from 27 (sieve) to 22 (Apex), contributing to the column height and cost reductions.

Pressure drop. Apex at 0.032 psi/tray and 0.697 psi total is the second-lowest after Dual-Flow (0.017 psi/tray, 0.543 psi total), and the lowest among trays with a downcomer. The low ΔP_{tray} comes from the combination of low β (0.38), low h_{weir} (0.5 in.), high f_{hole} (0.25) and fewer actual trays ($N = 22$).

Fouling resistance. Moving Valve and Ripple have the lowest open-area retention under fouling (0.70 and 0.72), predicting +104% and +93% dry-tray dP increases. Dual-Flow is the best of the conventional designs at 0.90 (+23%), followed by Fixed Valve (0.85, +38%). Apex at 0.93 (+16%) is the highest, reflecting PULSAR self-sweeping and the large-slot geometry.

Fabrication and cost. Apex’s relative unit cost factor of 3.20 is the highest (precision investment-cast cartridges plus helical sleeves plus AEGIS grid assembly). However, the installed cost is the lowest (0.64) because the shell dominates and Apex’s smaller diameter reduces shell cost by $\approx 49\%$.

12 Claims (Draft)

Claim 1. A vapour–liquid contacting tray comprising a structural cartridge-grid framework and an active deck panel, the active deck divided into a primary contacting zone ($\geq 60\%$ of the column cross-sectional area) comprising an array of axial swirl-tube cartridges, and a secondary contacting zone (10–20% of the column cross-sectional area) comprising lip-sealed dual-flow trickle valves, wherein the primary zone operates at a flooding capacity factor $k_{\text{cap}} \geq 1.8$ relative to the Fair sieve baseline at equal tray spacing, and the secondary zone sustains vapour–liquid contact at vapour loads below 10% of the primary-zone design basis.

Claim 2. The tray of Claim 1, wherein the swirl-tube cartridges each comprise tangential vapour-inlet slots (4–8 per tube), liquid-metering ports feeding from the deck pool below the slots, and a vane crown separator at the tube exit that discharges separated liquid into a collection annulus draining to the outlet-weir region of the same tray.

Claim 3. The tray of Claim 2, wherein the tangential slots are formed in a back-drivable helical sleeve that passively opens slot area with increasing vapour momentum, and the sleeve reverts to a fixed-geometry open position if mechanically seized.

Claim 4. The tray of Claim 1, wherein self-sweeping oscillator relief slots are located in the deck plate between cartridge footprints, each slot held closed by a spring or weight-loaded cap that opens momentarily when the deck differential pressure exceeds a set threshold and reseats thereafter.

Claim 5. The tray of Claim 1, wherein a variable-pitch cartridge layout and multi-chordal liquid-path baffles are applied across the primary zone to equalize the local vapour–liquid ratio, targeting a tray efficiency factor $\eta \geq 1.10$ above the O’Connell single-value baseline.

Claim 6. The tray of Claim 1, wherein the structural cartridge-grid comprises a welded support ring, deck panels locked to the ring by interlocking hook clamps, stiffening ribs at pitch $s \leq t\sqrt{\sigma_{\text{allow}}/(0.75q)}$, and integral relief flaps, providing a structural vapour-uplift rating of at least 21 kPa (3 psi) without reducing active bubbling area by more than 5%.

Claim 7. The tray of Claim 6, wherein the cartridges are individually removable through a column manway and replaceable without modification to the support ring or column shell.

Claim 8. The tray of Claim 1, having an active-area fraction f_{active} of 0.80–0.88, an open-area fraction f_{hole} of 0.20–0.30, and an outlet weir height of 0.25–1.0 inch.

Claim 9. A column containing trays according to any preceding claim.

Claim 10. A method of revamping a distillation column comprising replacing existing cross-flow trays with trays according to Claim 1 at equal or reduced tray spacing, thereby reducing column diameter requirements by at least 30% relative to a sieve tray sized on the same service basis.

13 Industrial Applicability

Capacity-constrained columns. The predicted 39% diameter reduction makes Apex applicable to shell-limited revamp situations where increasing throughput within an existing shell is the primary objective: C₃ splitters, demethanisers, FCC main fractionator top sections, and offshore columns where shell dimensions are fixed by deck loading or existing foundations.

Wide-range operating columns. The predicted 22.2:1 turndown makes Apex applicable to columns that must follow variable-rate feedstocks or operate at deep turndown without dumping; batch-integrated continuous columns, swing-service units, and grade-change-intensive operations.

Fouling and heavy-ends services. The 93% open-area retention under fouling and the PULSAR self-sweeping mechanism make Apex applicable to vacuum residue and atmospheric heavy-ends services where conventional trays foul within one run. The cartridge-swap maintenance procedure also reduces turnaround duration when complete tray replacement is needed.

Energy-intensive separations. The 54% reduction in total column pressure drop (General Rectification basis) reduces the reboiler duty at the base and condenser duty at the top, with a corresponding reduction in energy consumption per unit of separation. The efficiency improvement (+15 percentage points) reduces actual tray count and column height, providing capital savings and potentially allowing additional theoretical stages within the existing shell height.

Prior-art neighbours to clear. Shell ConSep/swirl-tube art and continuations; Koch-Glitsch SUPERFRAC and UFM multi-downcomer patents; lip-valve and check-valve tray art (Nutter, Koch, Sulzer); structural hold-down and bolt-down tray art; GRADEX/multi-chordal tray grading art (if separately patented); self-sweeping oscillator valve art. The asserted inventive step is the *integrated* architecture: pairing the primary swirl-tube zone with a dedicated secondary trickle zone for turndown, laying radial-grading and self-sweeping treatments over both, and mounting the result on a structural grid rated for 2–5 psi uplift, all in a single bolt-in deck.

14 Limitations and Model Caveats

1. **One-dimensional model.** The hydraulics engine (`column_hydraulics.py`) is a 1-D correlation framework. It cannot resolve the radial or azimuthal vapour/liquid distribution within a tray, the coupled behaviour of the two contacting zones, or the unsteady aerodynamics of the swirl-tube cartridges. The TrayType parameter set (APEX) is a parameterisation of the expected behaviour, not a simulation of the physical mechanism.
2. **Capacity factor** ($k_{\text{cap}} = 2.50$). This is the most influential single parameter (it controls all diameter results via Eq. (9)). It is derived from swirl-tube droplet-trajectory analysis and scaling arguments (Section 6.2), not from tray-scale measurements. Cold-flow and hot-rig confirmation is required before this value can be used in a final design.
3. **Minimum stable load fraction** ($f_{\text{min}} = 0.045$). This determines the 22.2:1 turndown claim and cannot be derived from a 1-D model. The two-zone mechanism (Section 6.3) is the proposed physical basis; its validity depends on LipSeal lip-sealing performance, sleeve-closure behaviour at low vapour rate, and absence of maldistribution between the two zones across the tray area.
4. **Service properties.** The General Rectification case is anchored to T801 stream data from `Hydraulics/HMB.xlsx`; liquid density, viscosity, surface tension and relative volatility are engineering estimates. The remaining four services use representative property sets. Substituting rigorous simulator-derived tray-level properties will shift all results.
5. **Cost model.** The two-proxy cost model (Section 6.7.11) is a screening tool, not an engineering estimate. It does not capture: detailed vessel mechanical design (nozzle re-rating, foundation loading), plot re-configuration, process-piping modifications, AEGIS welded-ring installation cost (one-time), cartridge procurement lead time, or vendor-specific fabrication pricing.
6. **Downcomer backup at High-P/High-L service.** Table 4 shows Apex at 54% downcomer backup for the High-Pressure/High-Liquid-Load service. Although within the 100% limit, this is the highest Apex figure across all services; detailed downcomer sizing with actual tray-level liquid loads should be conducted.
7. **Physical mechanisms not modelled.** The model does not capture: PULSAR slot cycling dynamics, GRADEX-driven local vapour/liquid equalisation, HT-08 sleeve response lag, or the interaction between the crown-separator liquid annulus and the deck pool at off-design vapour rates.

15 References and File Index

Hydraulics correlations.

- Fair, J.R., “How to Predict Sieve Tray Entrainment and Flooding,” *Petro/Chem Engineer*, 1961. [Capacity C_{SBF} correlation.]
- Kister, H.Z., *Distillation Design*, McGraw-Hill, 1992, Ch. 6. [Dry-tray orifice equation, Francis weir, downcomer apron loss, tray selection guidance.]
- O’Connell, H.E., “Plate Efficiency of Fractionating Columns and Absorbers,” *Trans. AIChE* 42 (1946). [Overall column efficiency E_{OC} .]
- Towler, G.P. and Sinnott, R.K., *Chemical Engineering Design*, 2nd ed., Elsevier, 2013. [Cost model component-weight rationale.]

Related HyTray disclosures.

- HT-01A/B/C (Hinge, Spring, LipSeal dual-flow valves) — secondary zone element source.
- HT-02 CVS (Centrifugal Vapour-Swirl tray) — primary zone base mechanism.
- HT-03 GRADEX (radial/azimuthal vapour-load grading) — efficiency-factor source.
- HT-05 PULSAR v2 (self-sweeping oscillator relief) — fouling-retention source.
- HT-07 AEGIS (surge-tolerant locked tray) — structural grid source.
- HT-08 VORTEXA (adaptive helical-sleeve swirl tray) — primary zone slot-modulation source.

Reproducible computation files.

- HyTrays/tray_library.py — TrayType dataclass and all seven tray parameter sets.
- HyTrays/column_hydraulics.py — hydraulics engine (Fair, orifice, Francis weir, downcomer, O'Connell).
- HyTrays/cost_model.py — first-order relative installed cost model.
- HyTrays/service_cases.py — five ColumnCase definitions.
- HyTrays/run_apex_paper.py — generates output/apex_paper_comparison.csv, output/apex_paper_table and Figures 3a–5.
- HyTrays/make_apex_drawings.py — generates Figures 1–6.